

Project Title: **SwapSphere** – A Smart Barter Platform with Intelligent Exchange Matching

1. Introduction

The increasing use of online marketplaces has made buying and selling goods more convenient than ever. However, many useful items remain unused because their owners no longer need them, while others may be searching for those same items. Existing platforms primarily focus on monetary transactions and provide limited support for direct item exchange. SwapSphere is a web-based platform designed to modernize the traditional barter system by enabling users to exchange items directly without involving money. The platform aims to provide a simple, secure, and intelligent environment for users to discover suitable exchange opportunities.

2. Problem Statement

Although barter is a practical and sustainable method of exchanging goods, finding suitable exchange partners remains difficult. Users often struggle to identify people who possess the desired item while also being interested in what they have to offer. In addition, the lack of trust between strangers discourages users from participating in direct exchanges. Existing marketplaces do not effectively address these challenges, making item-for-item exchange inconvenient and inefficient.

3. Objectives

- To develop a web-based platform for direct item-to-item exchange without monetary transactions.
- To enable users to create listings and specify multiple preferred exchange items.
- To implement a reputation system based on user verification, exchange history, and ratings to improve trust.
- To develop an intelligent exchange matching engine that recommends suitable exchange opportunities based on user preferences, item compatibility, estimated value, distance, and reputation.
- To support both direct (two-way) exchanges and automated three-way exchange cycles.
- To provide an Exchange Room where matched users can coordinate and complete exchanges.

4. Scope of the Project

The proposed system will be developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The platform will include Google Authentication, email login, user profile management, item listing, search and filtering, reputation management, exchange requests, intelligent matching, direct and three-way exchange workflows, Exchange Rooms, and ratings and reviews. Features such as online payments, courier integration, temporary lending, and four-way exchange cycles are excluded from the initial version and may be considered as future enhancements.

5. Expected Outcome

The proposed system is expected to provide a secure, user-friendly, and efficient platform for direct item exchange. By combining a transparent reputation system with an intelligent matching engine, SwapSphere will simplify the process of finding compatible exchange partners while reducing the limitations of traditional barter systems. The project is also

expected to encourage the reuse of existing resources, promote sustainable consumption, and demonstrate an innovative approach to modern item exchange through support for both direct and three-way exchanges.